

Bahram Behzadian

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Research Engineering | Reinforcement Learning | Evaluation & Post-Training Google Scholar | U.S. permanent resident

Profile

Research scientist combining reinforcement-learning research with production ML systems. At Meta, I build systems for experiment-metric correctness, anomaly and data-quality monitoring, and ML auditing in large-scale ads experimentation. My research spans robust MDPs and policy improvement under truncated evaluators with verifier-style rewards, with applications to RL post-training and reasoning.

Experience

2022–Present **Research Scientist**, *Meta*

- Build production systems for experiment-metric correctness and ML auditing in large-scale ads experimentation.
- Develop anomaly and data-quality monitoring together with statistical diagnostics for unreliable, biased, stale, or inconsistent signals.
- Translate failure analyses into monitoring and correctness safeguards for production experimentation systems.

2018 **AI Engineer Intern**, *Envio, Inc.*, Dover, NH

- Developed learning-based methods for vehicle routing and scheduling under operational constraints and uncertain execution.

Selected Research

Policy Improvement under Imperfect Evaluators

UPI-TRM, ICLR 2026 Workshop on AI with Recursive Self-Improvement

Derived finite-depth policy-improvement guarantees that separate Bellman residual from truncation bias, and studied policy improvement with verifier-style rewards under truncated evaluators.

Robust and Risk-Sensitive Markov Decision Processes

NeurIPS 2021 and AISTATS 2021

Developed scalable algorithms for L_∞ -constrained S-rectangular robust MDPs and percentile criteria under model uncertainty.

Selected Publications

UPI-TRM: Policy Improvement under Truncated Evaluators with Verifier-Style Rewards

Bahram Behzadian, Brett Daley, Gopeshh Subbaraj, Houssam Nassif

ICLR 2026 Workshop on AI with Recursive Self-Improvement [Paper]

Fast Algorithms for L_∞ -Constrained S-Rectangular Robust MDPs

Bahram Behzadian, Marek Petrik, Chin Pang Ho

Conference on Neural Information Processing Systems (NeurIPS), 2021 [Paper]

Optimizing Percentile Criterion using Robust MDPs

Bahram Behzadian, Reazul Hasan Russel, Marek Petrik, Chin Pang Ho

International Conference on Artificial Intelligence and Statistics (AISTATS), 2021 [Paper]

Fast Feature Selection for Linear Value Function Approximation

Bahram Behzadian, Soheil Gharatappeh, Marek Petrik

International Conference on Automated Planning and Scheduling (ICAPS), 2019 [Paper]

Monte Carlo Localization in Hand-Drawn Maps

Bahram Behzadian, Pratik Agarwal, Wolfram Burgard, Gian Diego Tipaldi

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2015 [Paper]

Complete publication list: [Publications page](#) | [Google Scholar](#)

Technical Skills

Programming: Python, PyTorch.

Research and ML: reinforcement learning, robust MDPs, policy improvement, model-based RL, partial observability, verifier-style rewards, RLVR.

Post-training: RLVR, verifier rewards; working knowledge of PPO, DPO, GRPO, reward modeling.

Education

2015–2022 **Ph.D. in Computer Science**, *University of New Hampshire*

Dissertation: *Efficient Data-Driven Robust Policies for Reinforcement Learning* (advisor: Marek Petrik).

M.S. in Computer Science awarded en route (2018). Thesis: *Representation Learning for Reinforcement Learning via Singular Value Decomposition*.

2010–2013 **M.S. in Machine Automation**, *Tampere University*

Thesis: *Robot Localization under Partial Observability with Weak Maps*

Honors and Fellowships

CEPS Graduate Fellowship, University of New Hampshire

Thesis and Dissertation Fellowship, Tampere University of Technology